



USER GUIDE

TRIMBLE MOBILE IMAGING

SOFTWARE

Revision L
April 2026

Legal Notices

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Legal Notices

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Release Note

This is the April 2026 release revision L of the **Trimble TMI Manual, P/N T001242**. It applies to:

- Trimble MX scan system (MX90, MX60, MX9, MX50)
- Trimble Mobile Mapping system MX7

The T001242 original document is written in English. All documents in other languages are translations from the original English document.

Document History

Date	Revision	Remarks
January 2019	A	First release
March 2019	B	Updates
December 2019	C	Updates
May 2021	D	MX50 updates
November 2021	E	External connector updates
May 2022	F	Trimble MX9, Dual Head, AP60, Spherical+3x12MP Support
November 2022	G	Added mission coverage upload and download functionality
March 2024	H	Trimble MX90
December 2024	J	Trimble MX60
July 2025	K	Updated Trimble Support contact information
April 2026	L	Updates

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General Information

The Trimble TMI software can be used with the following Trimble Mobile Mapping systems:

- MX90
- MX60
- MX9
- MX50
- MX7

The MX90, MX60, MX9, and MX50 are referred to as **Trimble MX scan systems**.

As this User Guide is a general User Guide for the Trimble TMI software, it may not be fully applicable to your specific system or system setup. Differences are explained in the course of the User Guide.

Prerequisites to System Setup and Operation

CAUTION – Before starting, please make sure all connections are secure. If you are using a Trimble MX scan system, make sure the recording SSDs are inserted and locked.

Before starting data capture, make sure the measurement parameters (e.g. sensor setup on the vehicle and capture parameters like laser pulse rate) have been set properly. The parameters you have defined may be saved as presets under user-defined names and can be selected when starting a mission later.

A standard operation workflow and an “In-the-field operation checklist” are described more in detail in the following guides:

- *Trimble MX90 Quick Start Guide*
- *Trimble MX60 Quick Start Guide*
- *Trimble MX9 Quick Start Guide*
- *Trimble MX50 Quick Start Guide*
- *Trimble MX7 Quick Start Guide*

WARNING – If you are using a Trimble MX scan system, the first time you turn it on, and then every time you start a data collection campaign in a different country, you must specify the name of this country when setting the WiFi access point. This is a mandatory prerequisite to ensure you comply with the regulations of this country in terms of WiFi use. To set the country of use, run the TMI software, from the menu select **System Administration > WiFi > MX90/MX60/MX9/MX50 WiFi Access Point** and then choose the country. For more information, [Wifi Setup, page 51](#).

Establishing Network Connection

WLAN Connection

When using a mobile device to control a Trimble MX system, a WLAN connection can be used. With the Trimble MX7, you need to connect the WLAN adapter to the power box.

To be able to access the system using WLAN, the tablet computer must connect to the WiFi access point hosted by the system.

MX Scan System WiFi Parameters

SSID	Trimble MX90/MX60/MX9/MX50 (<serial number>)
Password	Refer to the label on the control unit

MX7 WiFi Parameters

SSID	Trimble MX7 (<serial number>)
Password	Refer to the label on the power box

If the WiFi password has been lost, please contact Trimble Imaging Support and provide the serial number of your Trimble system.

Ethernet Connection

With Trimble MX scan system

Using a standard Ethernet cable (Patch Cable), connect your device to the LAN Port of the MX scan system control unit. Your device will acquire an IP address via DHCP.

With Trimble MX7

The Ethernet adapter needs to be connected to the power box.

Using a standard Ethernet cable (patch cable), connect your device to the Trimble MX7 Ethernet adapter port. Your device will acquire an IP address via DHCP.

To avoid routing problems, it is recommended to disable the WiFi adapter on the laptop when using Ethernet connected to Trimble MX7.

TMI

TMI (for Trimble Mobile Imaging) is the operating software for Trimble Mobile Mapping systems. It is a web-based application and therefore no separate or additional software installation is required. The TMI software consists of two modules:

- TMI.Capture (field operating software)
- TMI.AI (administration interface)

To start TMI.Capture in your browser, you need to connect your device (a tablet or laptop) to the Trimble Mobile Mapping system either via WiFi or via LAN cable.

Google Chrome is the recommended browser (other existing browsers have not been tested). Start TMI.Capture by typing the following web-address in your browser:

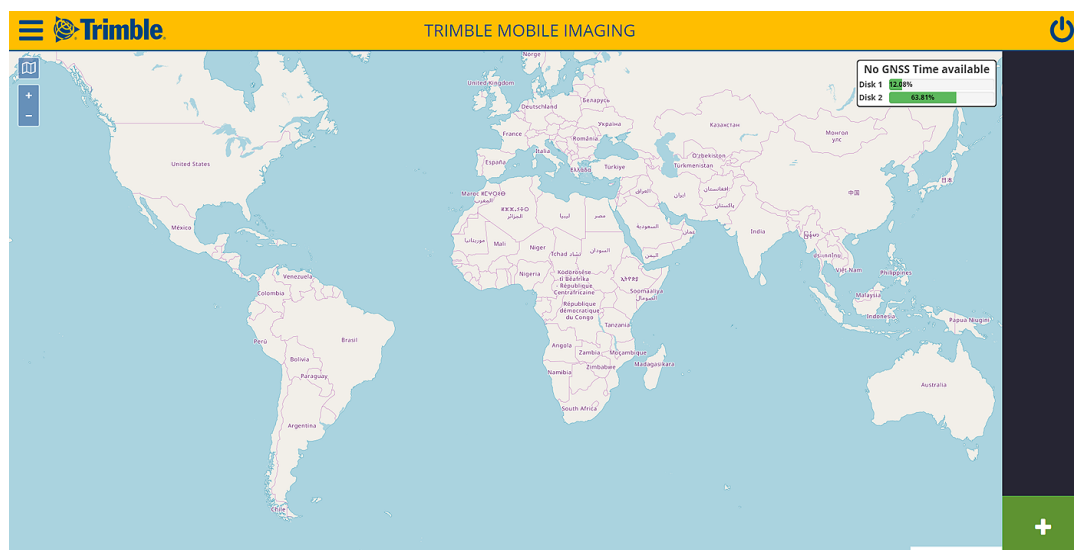
- With MX90: `http://tmi.mx-scan.net`
- With MX60: `http://tmi.mx-scan.net`
- With MX9: `http://tmi.mx9.net`
- With MX50: `http://tmi.mx50.net`
- With MX7:

Connected via WLAN (WiFi): `http://192.168.173.1/`

Connected via Ethernet: `http://192.168.174.1/`

To learn how to build the SSID of your Trimble Mobile Mapping system, see [WLAN Connection, page 5](#).

Just after system startup, the TMI.Capture main window comes up and looks like this:



The OSM map is shown in the map window only when the system is connected to the Internet.

Buttons and Color Status

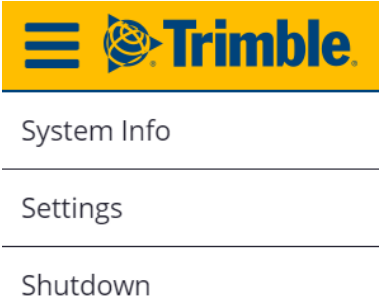
Icon	Function				
	Access to TMI.Capture main menu				
	Used to shut down the system				
	Used to close the mission in progress				
	Gives access to the Help menu				
	Access map layers				
	Map zoom in / zoom out button				
	Changes map orientation (North vs. vehicle)				
	Auto centers the map around the vehicle location				
	Opens the dashboard giving an overview on system information				
	Camera view Select one of the following devices: <table><tr><td>Front Left</td><td>Front Right</td><td>Back Down</td><td>360°</td></tr></table>	Front Left	Front Right	Back Down	360°
Front Left	Front Right	Back Down	360°		

- Color status:
- Red: Device error
 - Orange: Device is not time-synchronized yet
 - Green: Device time synchronization successfully completed

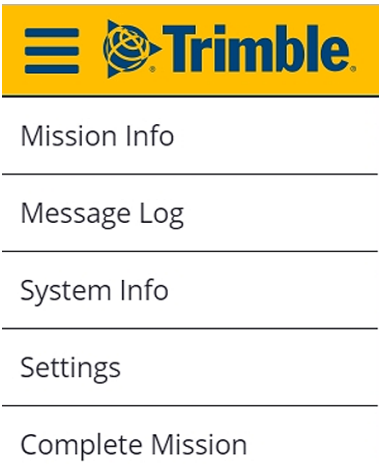
Icon	Function
	Navigation view Color status: • Red: No valid navigation solution is available or navigation system failure (no recording possible). • Orange: Navigation solution available (laser and image data recording is possible, pos data recording automatically starts). • Green: Navigation solution meets user accuracy requirements (see Settings, page 17).
	Laser view Select one of the following devices: Laser Left Laser Right Color status: • Red: Device error • Orange: Device is not time-synchronized yet • Green: Device time synchronization successfully completed
	Comments: Allows you to enter comments during a mission. All comments are time-tagged and saved to the database, according to the mission timeline
	Change capture settings: Allows you to change capture settings for both lasers and cameras.
	Recording button Color status
	• Red: Data logging is active • Green: No data logging
	Starts a mission.

Main Menu

The TMI.Capture menu contains the following options if no mission has been started yet:



The TMI.Capture menu contains the following options if a mission has already been started:



Map Layers

This function allows you to display different mission layers in the map window. Mission layers (also known as "mission coverage" layers) show capture information (trajectory, record status, etc.) of current and previous missions.

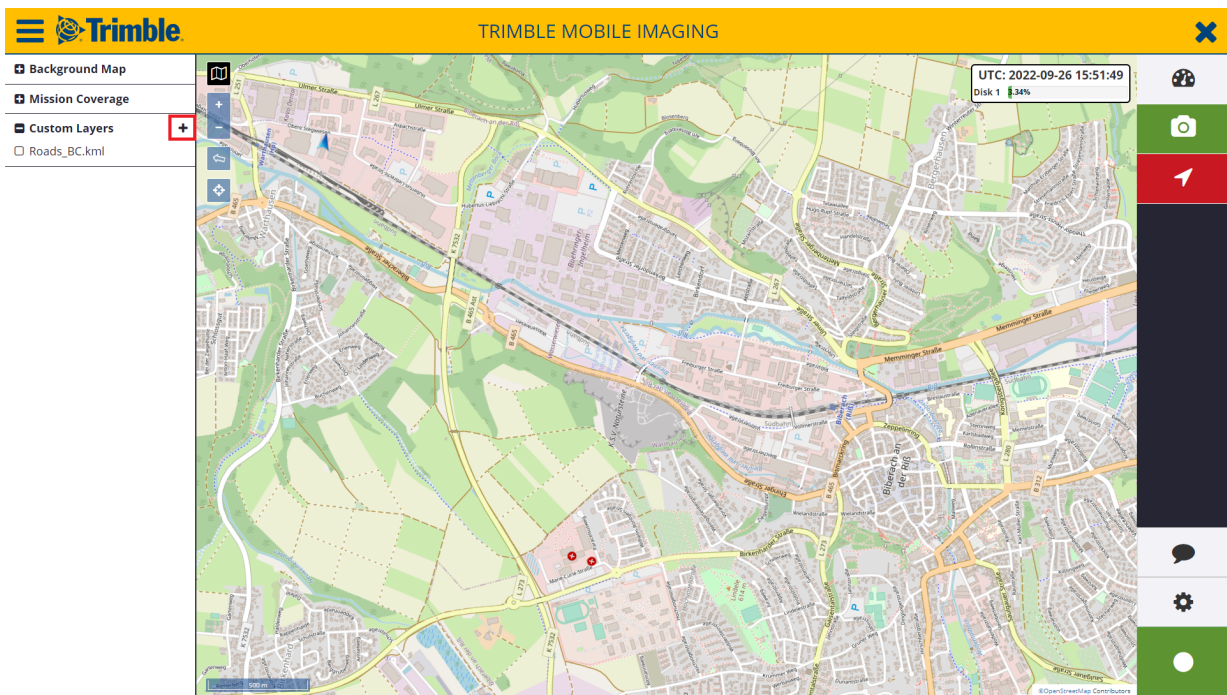
Custom layers can be uploaded as KML files and displayed in combination with mission layers.

Open List of Map Layers

1. Connect the tablet or computer to the system using WiFi or an Ethernet connection.
2. In the TMI.Capture map window, click  at the top left of the map to open the **Map Layers** menu.

KML Import

1. To add a KML file, click the + next to the **Custom Layers** option in the **Map Layers** menu.

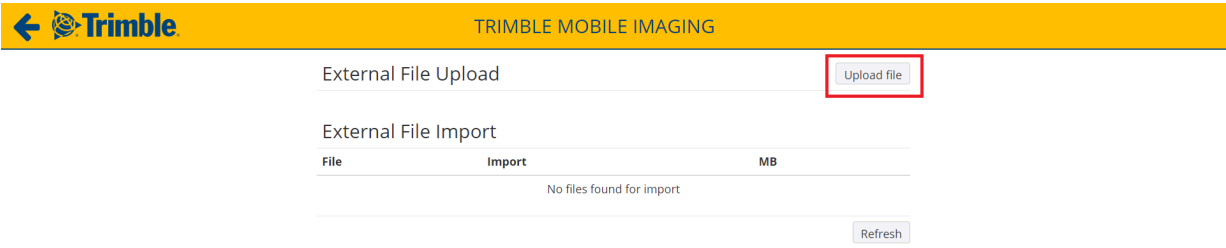


2. You can then choose to import the KML file using the browser or from a USB drive.

NOTE – The maximum KML file size that can be imported is currently limited to 5 MB.

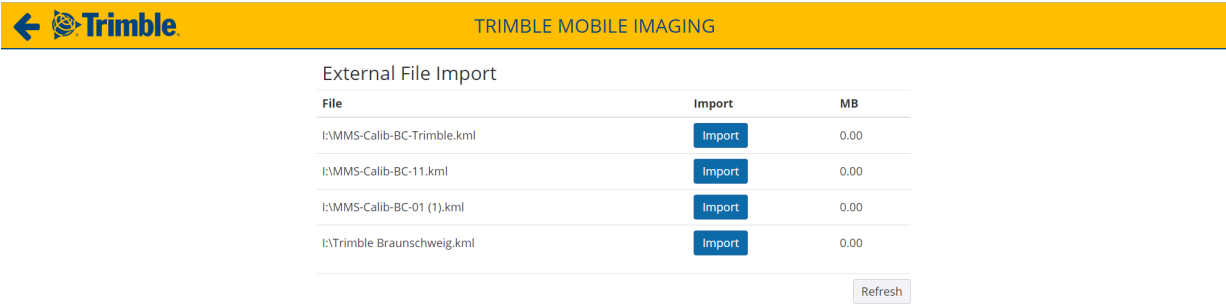
Import KML file via browser upload

1. Click **Upload file**.
2. Select the KML file.



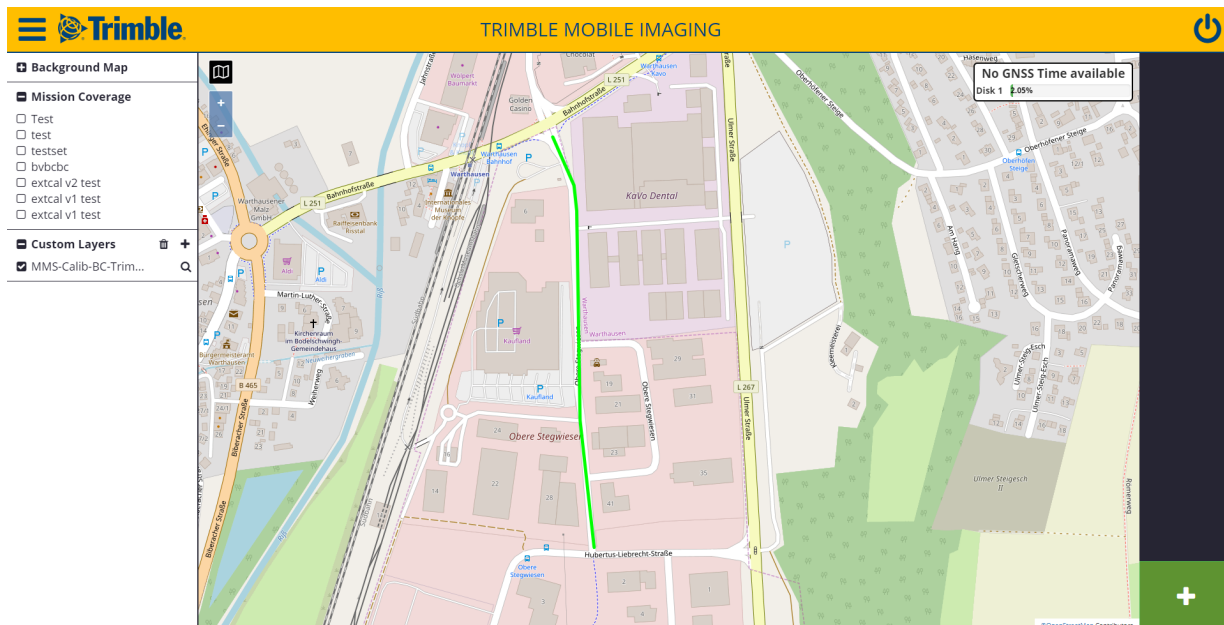
Import KML file from USB drive

1. Insert the USB drive (memory stick) that contains the KML files:
 - Into the USB 1 port on the control unit (MX scan system)
 - Into the USB port on the sensor unit or power unit (MX7)
2. Click **Refresh**.
3. Click the **Import** button next to the file name you want to import.



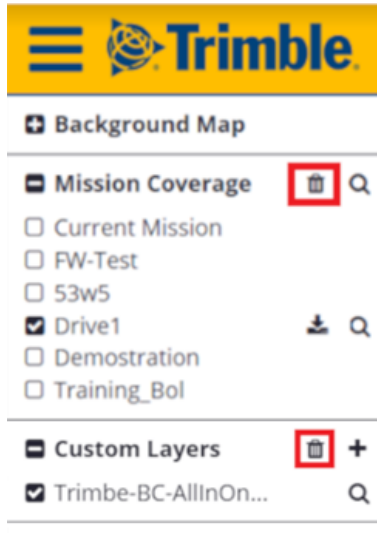
KML View

1. In the **Map Layers** menu, click **Custom Layers**.
2. Select the KML file you want to display.
3. Click the magnifying glass next to the file name to center the content of the KML file on the screen. Your kml file is shown on the screen.



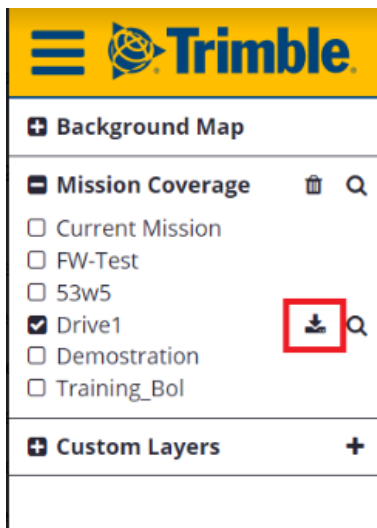
Delete a Map Layer

1. In the **Map Layers** menu, expand the **Mission Coverage** or **Custom Layers** menu.
2. Select the mission coverage or custom layer file.
3. Click the trash icon  next to the menu you have expanded to delete the selected file.



Export Mission Coverage

Click the **Download** button next to the active Mission Coverage to download a .kmz file which contains the trajectory of the selected mission.



Mission Info

This function provides a summary of the missions settings (only if a mission is active), including mission name, vehicle settings and capture settings for the mission.

Mission Info

Mission Name:	Test 2018-02-21-01
Capture:	Urban Area
Sensor.Enabled.Camera_1	true
Sensor.Enabled.Camera_2	true
Sensor.Enabled.Camera_3	true
Sensor.Enabled.Camera_4	true
Sensor.Enabled.Laser_1	true
Sensor.Enabled.Laser_2	true
Accur.Attitude	0.2
Accur.Heading	0.75
Accur.Position	10
Accur.Velocity	1
Camera.Trigger.Type	1
Camera.Trigger.Distance	3
Camera.Trigger.FrameRate	1
Laser.VQ.Preset.MeasProg	3
Laser.VQ.Preset.LineSpeed	250
Vehicle:	Default Vehicle
System.MountHeightAboveGround	2
DMI.Enabled	false
GAMS.Enabled	false

Message Log

This function provides an overview of the logged system information. Data can be sorted by user name. A data filter can also be added: **All** for listing all events (i.e. no filter), **Alarm** for listing only the alarm events, or **Warning** for listing only the warning events.

 TRIMBLE MOBILE IMAGING		
Module	Message	Timestamp
System	Mission started ID TMX9117070101-000889	2018-02-21 10:42:11UTC
System	Client Connected	2018-02-21 10:41:55UTC

All
Alarm
Warning

First
<
1
>
Last

System Info

This function provides identification information on the Trimble Mobile Mapping system used.

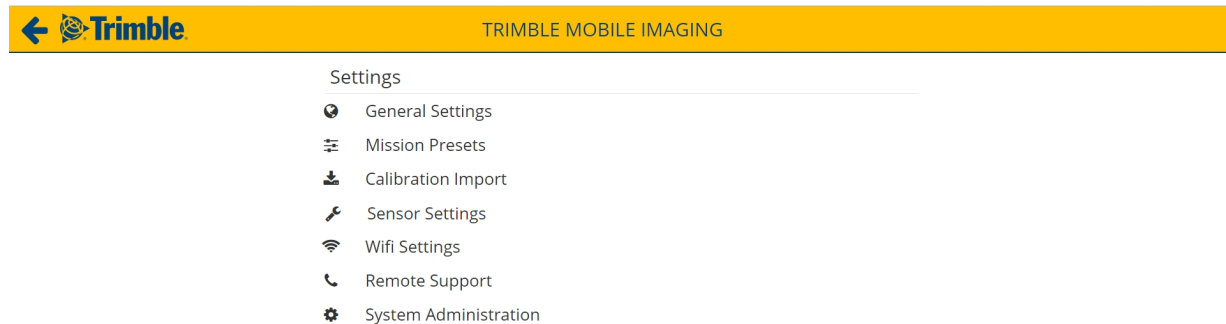
 TRIMBLE MOBILE IMAGING	
About	
Manufacturer	Trimble
Product Name	MX9
Serial Number	117070101
Asset Tag	TMX9117070101
Software Name	Field Software Update
Software Version	0.8.2
Maintenance Expiration Date	2019-02-16
Days Remaining before Maintenance expires	360

Settings

This function allows you to make the following changes to your Trimble Mobile Mapping system:

- Define or change several settings for data capture (presets).
- Define how to use the TMI software.

When selecting **Settings**, the following sub-menu appears:



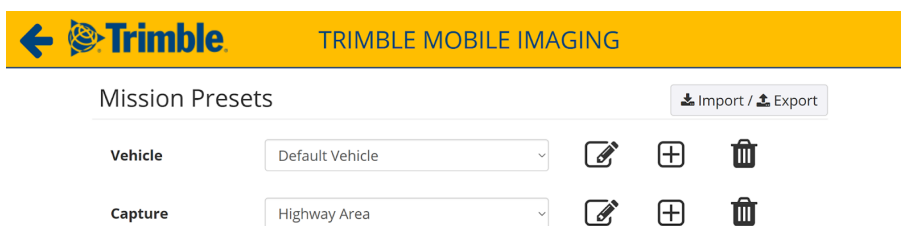
- **General Settings**

[Language]: Allows you to choose the interface language (if language packages are available).

- **Mission Presets**

Allows you to define various presets (i.e. default values) for Vehicle and Capture settings. These presets will be used to define the settings when starting a mission. However you can easily change the Capture settings during a mission by doing a Mission re-configuration.

-  Allows you to edit the selected preset.
-  Allows you to create a new preset.
-  Allows you to delete the selected preset.



Import / Export of Mission Presets

All vehicle and capture presets can be exported together to a single file. Exported preset files can be saved to a USB drive or downloaded via the browser. This is useful for backing up your complete preset configurations or for transferring them to another Trimble Mobile Mapping system.

You can import all previously exported preset files to restore or transfer all vehicle and capture presets at once. You can import preset files from a USB drive or upload them using the browser. When importing, you can choose between two options:

- **Overwrite:** Replaces all existing presets with the imported ones. Use this option to fully restore a previous configuration or to replicate the exact setup from another system.
- **Append:** Adds the imported presets to the existing ones. Use this option to merge presets from another system without losing your current configurations.

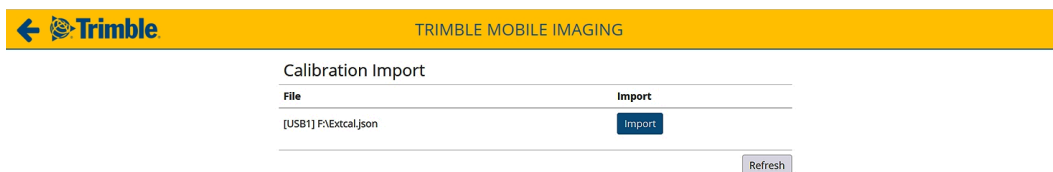
NOTE – When importing presets, the system will verify that the presets are compatible with the connected Trimble Mobile Mapping system. If the presets were created for a different system type, some settings may not be applicable and import will fail.

• Calibration Import

You can import new boresight calibration values into your Trimble Mobile Mapping system. After a boresight calibration has been made, a new calibration file can be used to update the settings of the system. The calibration file is also known as the JSON file in this user guide.

Below is how you can update the system settings:

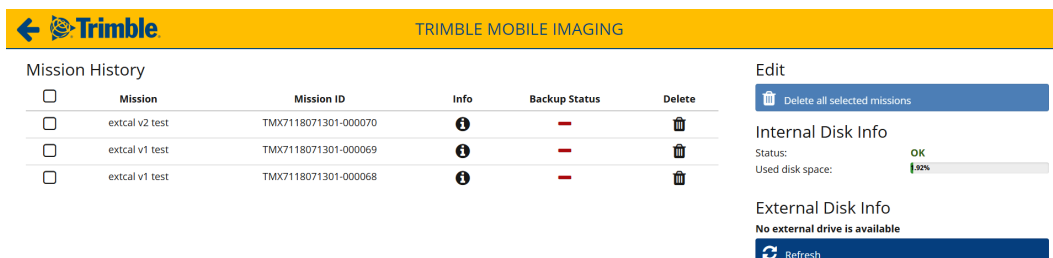
- Copy the calibration file (JSON file) to a USB memory stick.
- Plug the USB memory stick to the USB1 socket on the control unit and press the Import button next to the file name.



• Backup (MX7 only)

The Backup Data View provides information about data located on the Trimble MX7 internal disk as well as information about the status of the internal disk. Status of the external disk is shown only when the external disk is connected to the USB 3.0 port on the MX7 sensor head (not on the USB port of the power supply).

Within this menu, mission data located on the internal disk can now be copied to an external disk.



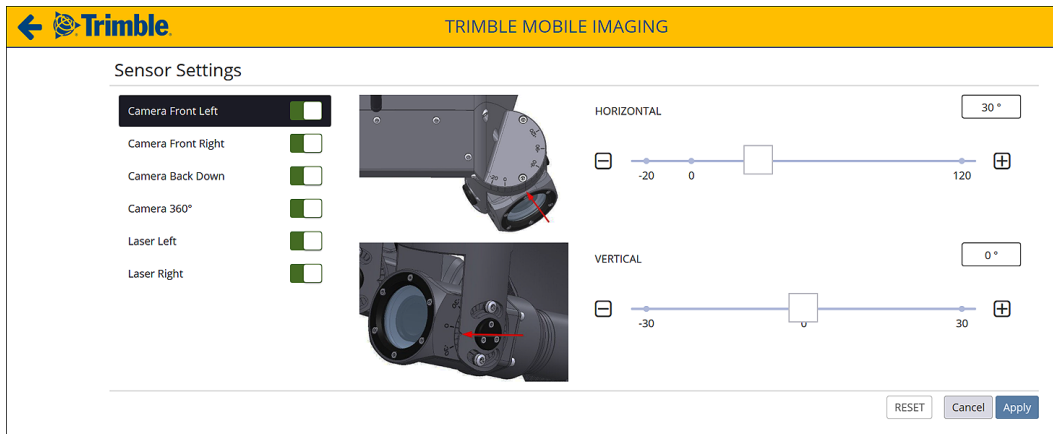
On the right side of the backup screen is information about the internal disk and external disks.

The **Mission History** section includes all missions that are currently stored on the internal disk of the Trimble MX7 System. For each mission, information about the Mission Name, Mission ID and Backup Status is shown. By clicking the info icon, more information is shown about the selected mission.

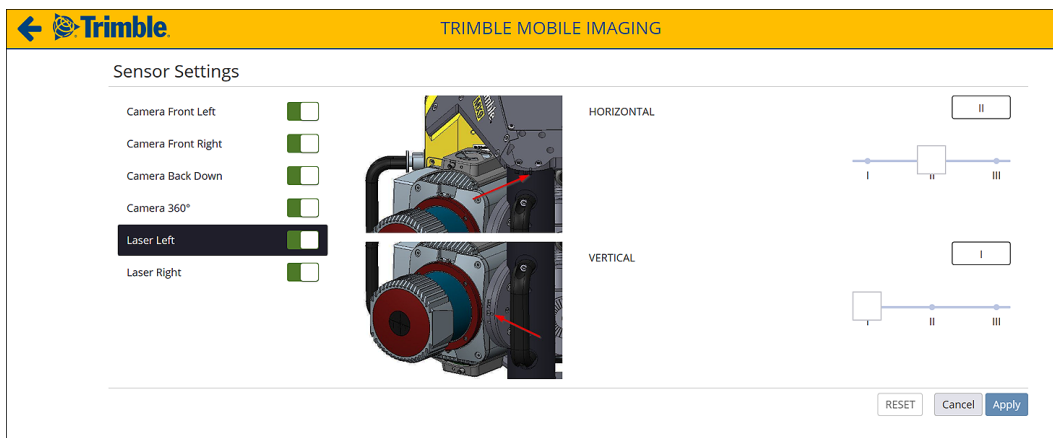
- **Sensor Settings (MX90/MX9 only)**

Once you have changed the mechanical orientation of the sensor components, you have to set up the MX90/MX9 system with the new orientation values. You can also enable or disable specific sensor components.

Camera, front left:



Laser, left:



CAUTION – A sensor will not collect any data if you disable it in this menu and it CANNOT be re-enabled while running a mission (even though the capture settings for this sensor may have been properly set).

CAUTION – The values you enter at this point are only "good start" values, which the system can use for the first mission (usually the calibration mission)..

Later in the office, those orientation values (for laser and/or cameras) need to be refined by a special processing called "boresight calibration". The results of this boresight calibration process (saved to a so-called JSON file) should then be provided back to the system **before** you start the next new missions. This is achieved by importing the "new" JSON file using a USB memory stick.

See also [Settings, page 17](#) (Calibration Import).

- **WiFi Settings:** Provides access to the TMI-AI software. For more details, see [Administration Interface Software, page 46](#).
- **Remote Support:** Provides access to the TMI-AI software. For more details, see [Administration Interface Software, page 46](#).

Mission Presets - Vehicle Settings

Set the parameters describing the mounting of the sensor unit on the vehicle. In this dialog, additional aiding navigation sensors, like a DMI or a secondary GNSS antenna (GAMS), can also be set up, and made active for data collection. Do not forget to use the **Save** button to save all the entered values.

Edit Vehicle Settings

Name:

Description:

Installation Parameter

Installation Height:

DMI

☐

Use DMI

GAMS

☐

Use GAMS

External Connector

☐

Use External Connector

Cancel

Save

CAUTION – If you do not activate an aiding navigation sensor here, the data from this sensor will NOT be logged during the mission, even though the required connections to the sensor unit may have all been made properly.

- **Name:** Enter a name for your vehicle setup (required). This name will appear later in the pull-down menu that you will open to select a vehicle preset from the list of existing ones.
- **Description:** You may add a description to this particular setup (optional).
- **Install Height:** Enter the installation height of the sensor unit on the car you are currently using. The installation height is measured directly from the external reference point on the MX Scan System Roof Rack to the road surface (the measured value is expressed in meters). For more details, see *Roof Rack Description and Installation* in the *User Guide* of the system you are using. This information will be essential at a later time for the post-processing software to be able to deliver accurate images to the viewer.
- **Use DMI:** *Only if a DMI is used*, activate this check button, assuming the DMI has been connected properly and you want to enable data logging. Check/enter the lever arm's XYZ values (in meters) and the mounting position (left or right, taking the driving direction as reference). Measure the wheel diameter with great care. For more details, see also *External Reference* section in the *Operations* chapter of the *Trimble MX7-DMI Installation & Operation Manual*.

DMI

☒ Use DMI

Lever Arm X:

0.5

Lever Arm Y:

-1.1

Lever Arm Z:

1.7

Wheel Diameter:

0.62

☒ Position Left
☐ Position Right

- **Use GAMS:** *Only if a GAMS is used*, activate this check button, assuming the GAMS has been connected properly and you want to enable data logging. Check/enter the lever arm's XYZ values (in meters). For more details, see also *External Reference Point* section in the *Operations* chapter of the *Trimble GAMS Antenna Kit Installation & Operation Manual*.

GAMS

☒ Use GAMS

Lever Arm X:

Lever Arm Y:

Lever Arm Z:

Cancel

Save

- **Use External Connector:** If an external signal connector is used, activate this check box.

NOTE – The available controls depend on the system type and may vary.

External Connector

☒ Use External Connector

COM Port

☐ Enable

Trigger 1

Signal Edge:

Rising Edge

Trigger 2

Signal Edge:

Rising Edge

Event 3

Signal Edge:

Falling Edge

Guard Time:

10

Event 4

Signal Edge:

Rising Edge

Guard Time:

10

- **COM Port Enable:** If a COM Port is used, activate this check box. There are options to change the **Baud Rate** and the **Update Rate** and to define which **NMEA Messages** should be used. For detailed information, refer to the *User Guide* for your MX system.

COM Port

☒ Enable

Baud Rate:

9600

Update Rate:

1 Hz

NMEA Messages

☐	GST
☐	GGA
☐	HDT
☐	ZDA
☐	VTG
☐	PASHR
☐	GGA2
☒	PPS
☐	GGK
☐	RMC
☐	EVT3
☐	EVT4

- **Trigger 1:** (MX90/MX60/MX9/MX50) Set up the Signal Edge to Falling or Rising.
- **Trigger 2:** (MX9/MX50) Set up the Signal Edge to Falling or Rising.
- **Event/Event 3:** (MX90/MX60/MX9/MX50/MX7) Set up the Signal Edge to Falling or Rising and Guard Time in msec.
- **Event 4:** (MX9/MX50) Set up the Signal Edge to Falling or Rising and Guard Time in msec.

TIP – For detailed information about the external connector, refer to the **External-Signal Connector** section of the *User Guide* for your MX system.

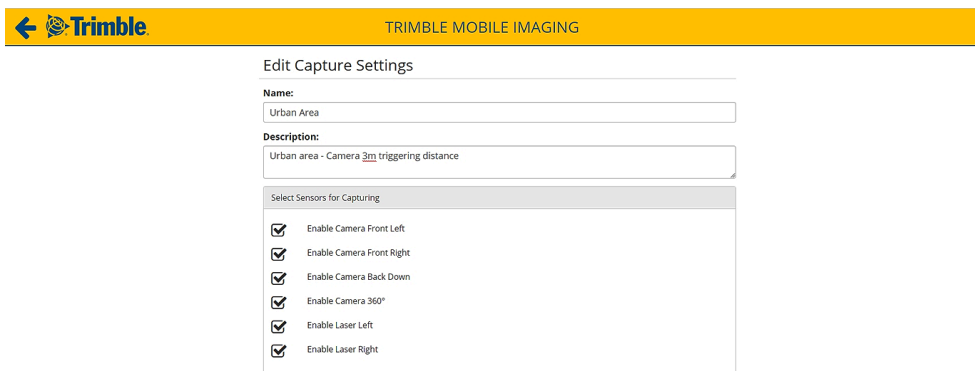
Mission Presets - Capture Settings

Use the **Capture Settings** dialog to do the following:

- Set each sensor (lasers, cameras) individually as an active or idle device
- Set the user accuracy figures for the navigation device
- Set the trigger settings for the camera and laser sensors.

You need to define the following parameters:

- **Name:** Enter a name for your capture setup. This name will appear later in the pull-down menu that you will open to select a vehicle preset from the list of existing ones.
- **Description:** You may add a description to this particular setup.
- **Select Sensor for Capturing:** You may activate or deactivate data logging for each sensor individually. The navigation sensor is always active.



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Edit Capture Settings

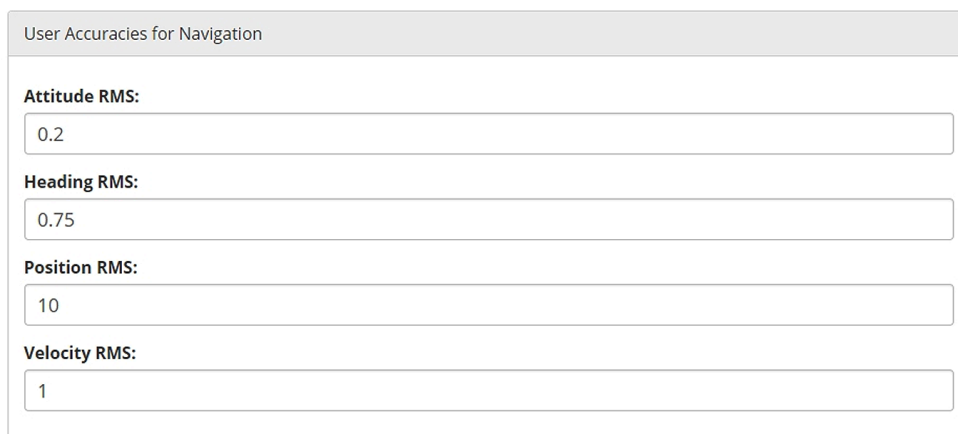
Name:
Urban Area

Description:
Urban area - Camera 3m triggering distance

Select Sensors for Capturing

- ☒ Enable Camera Front Left
- ☒ Enable Camera Front Right
- ☒ Enable Camera Back Down
- ☒ Enable Camera 360°
- ☒ Enable Laser Left
- ☒ Enable Laser Right

- **User Accuracies for Navigation:** These values indicate when the navigation button on the main page changes its status from orange to green.



User Accuracies for Navigation

Attitude RMS:
0.2

Heading RMS:
0.75

Position RMS:
10

Velocity RMS:
1

CAUTION – Changing the default values can ONLY be done by advanced users. The values you enter here must stay in compliance with the navigation system used by your Trimble Mobile Mapping system.

MX90 Default Figures:

Parameter	Description	Unit	Factory Default Value
Attitude RMS	Roll/Pitch value	degrees (°)	0.045
Heading RMS	Heading value	degrees (°)	0.060
Position RMS	XYZ position values	meters	10.000
Velocity RMS	Speed value	meters/second	0.025

MX60 Premium Figures:

Parameter	Description	Unit	Factory Default Value
Attitude RMS	Roll/Pitch value	degrees (°)	0.060
Heading RMS	Heading value	degrees (°)	0.045
Position RMS	XYZ position values	meters	10.000
Velocity RMS	Speed value	meters/second	0.025

MX60 Core/Pro Figures:

Parameter	Description	Unit	Factory Default Value
Attitude RMS	Roll/Pitch value	degrees (°)	0.075
Heading RMS	Heading value	degrees (°)	0.045
Position RMS	XYZ position values	meters	10.000
Velocity RMS	Speed value	meters/second	0.030

MX9 Default Figures:

Parameter	Description	Unit	Factory Default Value
Attitude RMS	Roll/Pitch value	degrees (°)	0.030
Heading RMS	Heading value	degrees (°)	0.300
Position RMS	XYZ position values	meters	10.00
Velocity RMS	Speed value	meters/second	0.300

MX50 Default Figures:

Parameter	Description	Unit	Factory Default Value
Attitude RMS	Roll/Pitch value	Degrees (°)	0.070
Heading RMS	Heading value	Degrees (°)	0.750
Position RMS	XYZ position values	meters	10.000
Velocity RMS	Speed value	meters/second	0.250

MX7 Default Figures:

Parameter	Description	Unit	Factory Default Value
Attitude RMS	Roll/Pitch value	degrees (°)	0.200
Heading RMS	Heading value	degrees (°)	0.750
Position RMS	XYZ position values	meters	10.000
Velocity RMS	Speed value	meters/second	1.000

- Trigger Settings for each Camera (MX7/MX50):**

Trigger Settings for each Camera

☒ Distance Based
☐ Fixed Frame Rate

Trigger Distance:

3

- **Distance Based:** Camera images are captured at constant driving distance, in meters. For the MX7, it will also set up the External Trigger.
- **Fixed Frame Rate:** Camera images are captured at constant frame rate (fps for "frames per second"). For the MX7, the External Trigger will not be triggered.
- **Trigger Distance or Trigger rate]:** Enter the measurement interval or distance.

- **Trigger Settings for each Camera (MX90/MX9):**

Camera Trigger Settings

Panorama Camera

☐ Distance Based
 ☒ Fixed Frame Rate

Frame Rate:

1.0 fps

Planar Cameras

☐ Distance Based
 ☒ Fixed Frame Rate

Frame Rate:

2.0 fps

Back Down Camera

☐ Distance Based
 ☒ Fixed Frame Rate

Frame Rate:

5.0 fps

- **Distance Based:** Camera images are captured at constant driving distance, in meters.
- **Fixed Frame Rate:** Camera images are captured at constant frame rate (fps for "frames per second").
- **Trigger Distance or Trigger rate]:** Enter the measurement interval or distance.
- **Settings for each Laser Scanner (MX90/MX9):**

Settings for each Laser Scanner

Measurement Program:

3 (1000kHz)

Line Speed:

250

Cancel

Save

NOTE – Values for “measurements/second” in the user interface are rounded values.

- **Measurement Program:** Measurements/second.
- **Line Speed:** Rotation speed of the scanner mirror (revolutions per second).

- Settings for each Laser Scanner (MX60):

Settings for each Laser Scanner

Laser Mode:

- Mode 1 (500kHz pulse rate, 120Hz line speed)
- Mode 2 (500kHz pulse rate, 200Hz line speed)
- Mode 3 (1000kHz pulse rate, 120Hz line speed)**
- Mode 4 (1000kHz pulse rate, 200Hz line speed)

External Connector (if enabled in Vehicle Setup)

NOTE – Values for “measurements/second” in the user interface are rounded values.

- **Laser Mode:** Combination of Measurements/second and Rotations/second.
- **Enable Dust Filter:** Enable the dust filter to eliminate dust above the road surface.
- **Lateral Range Limit:** Limits the laser range perpendicular to the driving direction. When enabled, laser points beyond the specified distance are discarded. The value is defined in meters and can be set between 5 m and 50 m. This setting applies to both laser scanners.

CAUTION – To avoid loss of data, the installation height must be set correctly in the **Vehicle Settings** screen.

Settings for each Laser Scanner

Laser Mode:

Mode 4 (1000kHz pulse rate, 200Hz line speed)

☒ Enable Dust Filter

Make sure the correct Installation Height is entered in the Vehicle Settings!

☒ Enable Lateral Range Limit

Maximum Distance:

20

Unit: m
Min 5 Max 50 Decimal Places 0

- Settings for each Laser Scanner (MX50):

Settings for each Laser Scanner

Pulse Rate:

500 kHz

☒ Enable Lateral Range Limit

Maximum Distance:

20

NOTE – Values for “measurements/second” in the user interface are rounded values.

- **Pulse Rate:** Measurements/second.
- **Lateral Range Limit:** Limits the laser range perpendicular to the driving direction. When enabled, laser points beyond the specified distance are discarded. The value is defined in meters and can be set between 5 m and 50 m. This setting applies to both laser scanners.
- **External Connector (MX90/MX60/MX9/MX50):**

NOTE – The available controls depend on the system type and may vary.

External Connector (if enabled in Vehicle Preset)

Trigger 1

☐ None
 ☒ Distance Based
 ☐ Fixed Frame Rate

Distance:

1

Trigger 2

☐ None
 ☐ Distance Based
 ☒ Fixed Frame Rate

Frame Rate:

6.6 fps

Trigger 1 and Trigger 2:

- **None:** Disables the External Trigger.
- **Distance Based:** External Trigger is triggered at a constant driving distance, specified in meters.
- **Fixed Frame Rate:** External Trigger is triggered by a constant frame rate, specified in fps (frames per second).

NOTE – External Trigger settings are applied only if the **External Connector** check box is enabled in the **Vehicle Presets** settings. See [Mission Presets - Vehicle Settings, page 20](#).

Complete Mission

Selecting this menu option immediately stops the mission in progress. Navigation data logging also stops immediately.

CAUTION – Navigation data logging also stops immediately when the mission is completed.

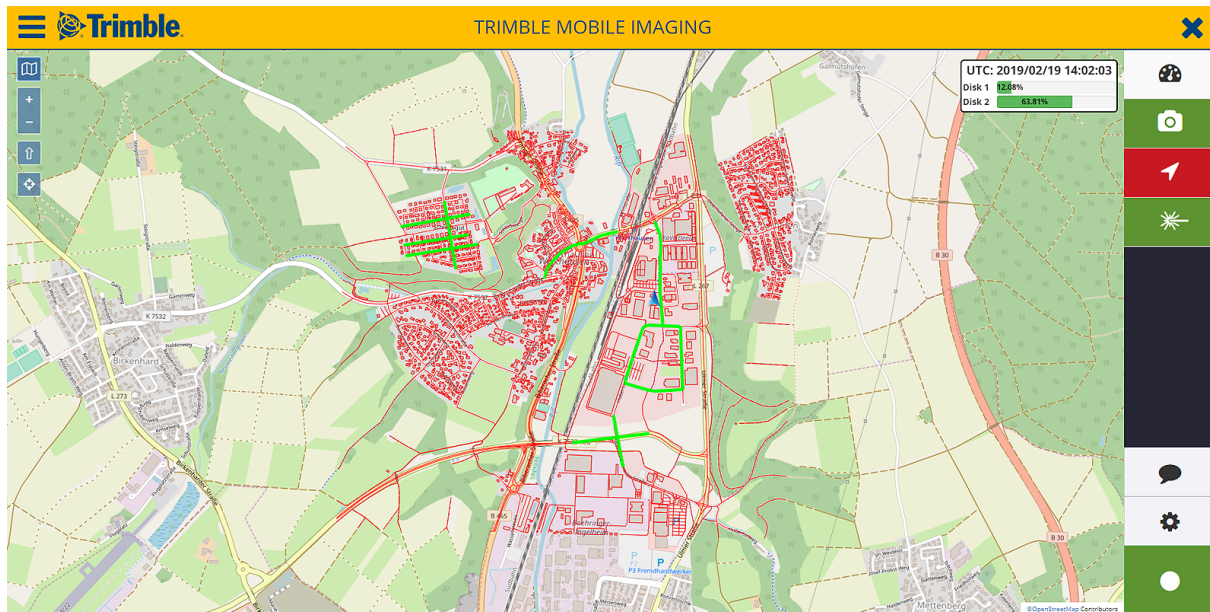
Shutdown

Selecting this menu option will shut down the system: All sensors AND the control unit will be powered off.

WARNING – To avoid the possibility of corrupted data, do not power off the Trimble Mobile Mapping system by simply removing power. Always power off the system using the **Complete Mission** AND **Shutdown** buttons.

TMI Map Window

The map window looks like this when a mission is started:

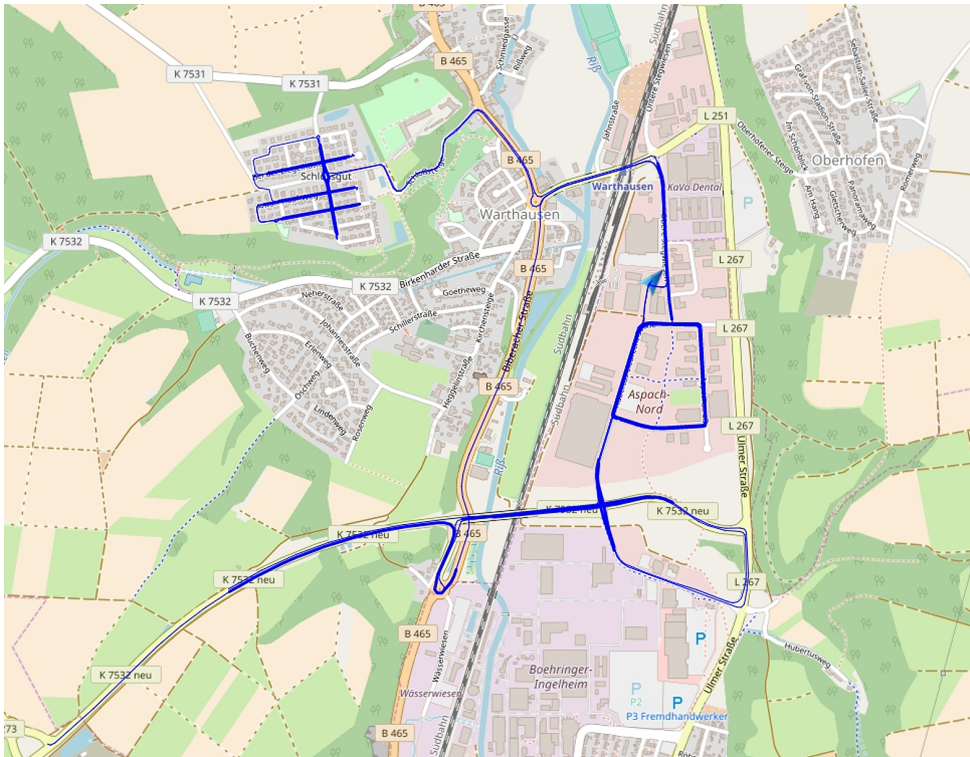


NOTE – The OSM map layer is visible only if the system has an active Internet connection.

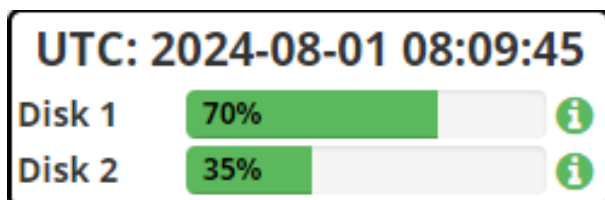
The map window provides the following functions (see also [TMI, page 7](#)):

- Access map layers
- Zooming
- Map orientation (North or vehicle steering direction)
- Auto-center around current vehicle location.

The trajectory is shown on the map as a thin blue line. Areas where data was recorded are shown with a thick blue line:

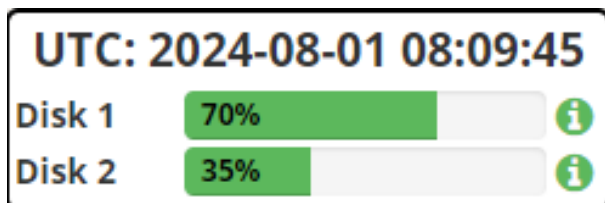


The picture above uses OSM data as background information for the map window. In the upper-right corner of the map window are the current UTC time and SSD fill-in status:



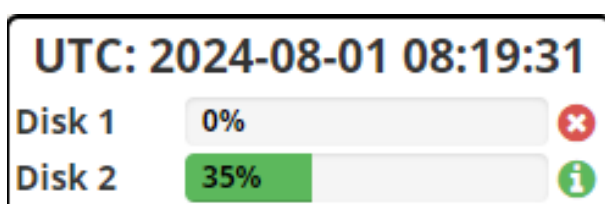
UTC time will appear automatically when GNSS initialization is complete.

At system startup and after each mission, data disks are checked. The result will be displayed as following:



If everything is fine, no icon will appear.

If there is a problem with a disk, this will be displayed:



There are 2 issue states.

- **Warning:** Mission can still run but data loss may occur under high system load.
- **Error:** The performance of the disk is compromised and proceeding with the mission may lead to data loss. If you wish to continue with the mission, confirm that the mission can start.

By clicking on the icons a detailed information about the issue is displayed.

TMI Dashboard

Press this button to open the Dashboard window:



The dashboard provides the following information:

- Navigation status
- Thumbnail views of panoramic camera 4 (if sensor currently active)
- Thumbnail views of oblique cameras 1, 2 and 3 (if sensor currently active).

Click anywhere within the map window to close this view mode.

TMI Camera View

Press this button to open the Camera View:



The Camera View is shared by the four cameras. To select which camera output to view on the Camera View, click on the corresponding button:



- **Front Left:** Oblique camera No. 1 (MX90/MX9 only)
- **Front Right:** Oblique camera No. 2 (MX90/MX9 only)
- **Back Down:** Oblique camera No. 3 (MX90/MX9 only)
- **360°:** Panoramic camera

For each camera, the Camera View shows a real-time thumbnail preview of the actually recorded image. Additionally, some settings can be made in real time for each camera.

Click anywhere within the map window to close this view mode.

Panoramic Camera

Using the **Auto** button or the slider, you can change the settings of the panoramic camera. Using the slider, you can change a mixture of exposure time and gain:

- Slider to the left: Image gets darker
- Slider to the right: Image gets brighter

Oblique Cameras (MX90/MX9/MX60 only)

You will get the same type of view for camera No.1 (front-left), camera No. 2 (front-right) and camera No. 3 (back-down).

NOTE – The MX60 has the back-down camera only.

Using the slider, you can change the exposure compensation for each oblique camera:

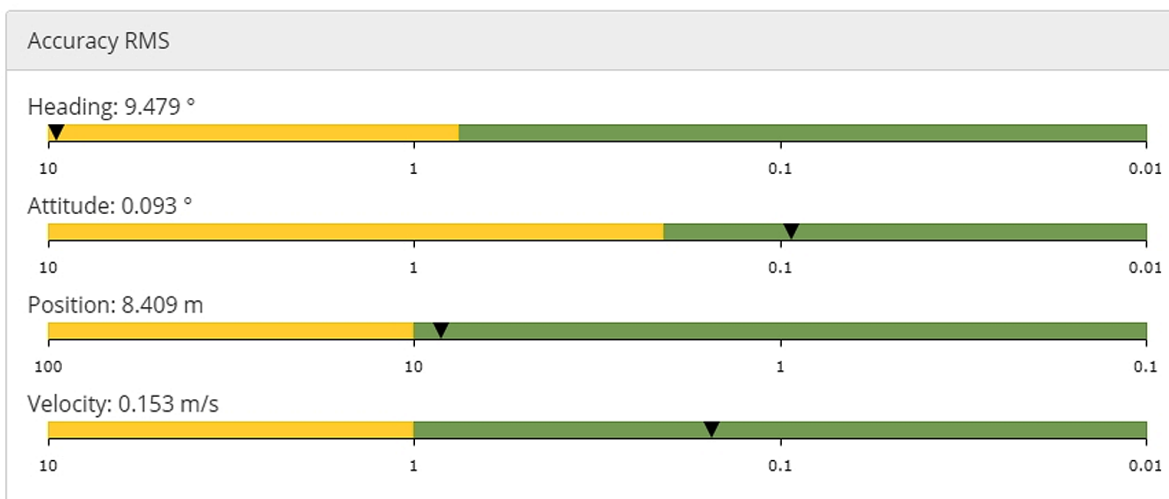
- Slider to the left: Image gets darker
- Slider to the right: Image gets brighter

TMI Navigation View

Press this button to open the Navigation View:



The Navigation View provides detailed information about the accuracy of your navigation system. Below is an example of the reported information:

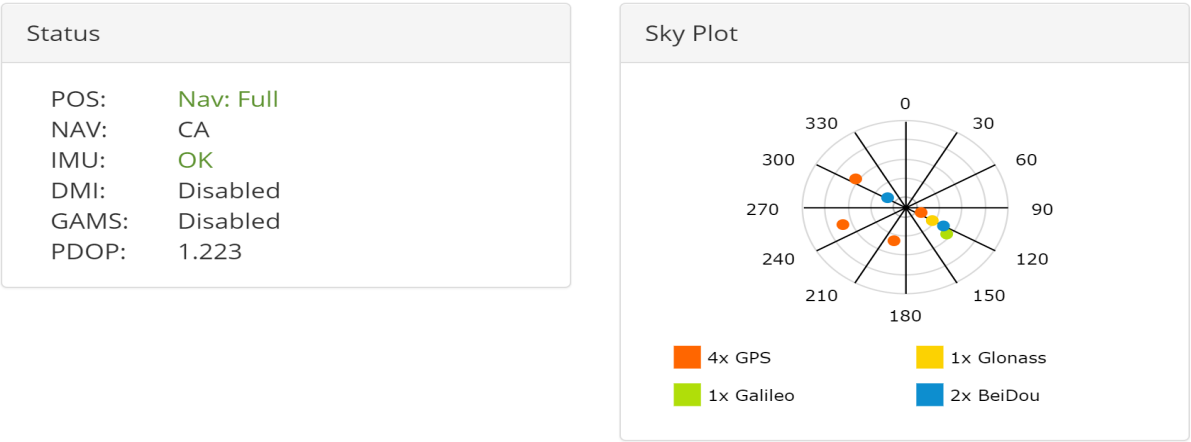


This view uses a logarithmic scale. The values for which the color changes from orange to green are defined in the **User Accuracies for Navigation** section of the Capture Settings (see [Mission Presets - Capture Settings](#), page 25).

CAUTION – Be aware of the following:

- Navigation data is logged as soon as a valid position is available and the blue arrow is visible on the main map window.
- Navigation data logging is NOT affected by the status color changing from orange to green or vice versa.
- Navigation data logging stops as soon as a mission is closed.

Additional status information and a satellite skyplot are available on this view as well. See example below.



Click anywhere within the map window to close this view mode.

TMI Laser View (MX scan system)

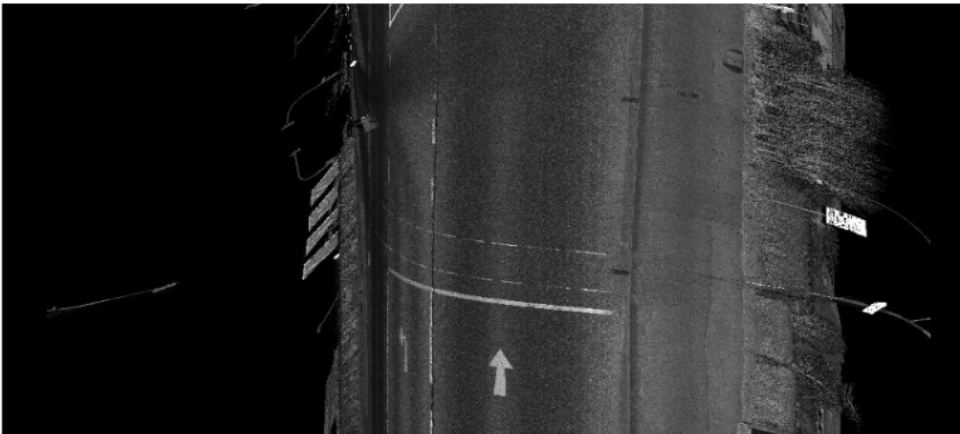
Press this button to open the Laser View (only available if at least one laser is currently active):



The Laser View is shared by the two laser scanners. To select which laser output to view on the Laser View, click on the corresponding button:



The view provides a reduced real-time data stream and some other laser related information. The waterfall view is shown below:



Laser Information	
Name	Value
GPS	✓ SYNCED_WITHIN_LAST_SECOND
Laser PRR	500 kHz
Line Speed	118.843 Hz

Click anywhere within the map window to close this view mode.

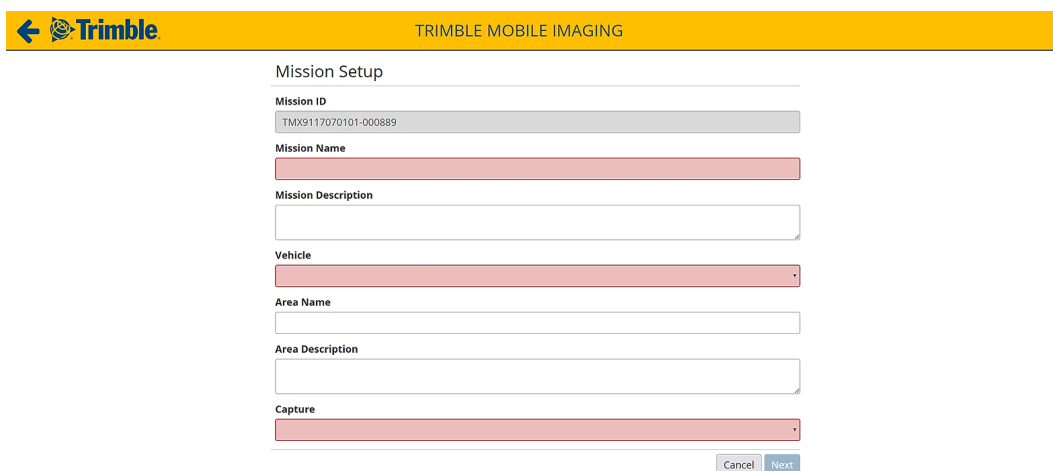
Collecting Data

Starting a Mission

1. In the main window, press  to start a mission.

NOTE – This button is disabled while a disk check is still in progress (Initializing/Analyzing). Wait until the disk check is complete before starting a mission.

The following menu appears:



← Trimble TRIMBLE MOBILE IMAGING

Mission Setup

Mission ID
TMX9117070101-000889

Mission Name

Mission Description

Vehicle

Area Name

Area Description

Capture

Cancel Next

You need to define, at least:

- Mission name
- Vehicle settings
- Capture settings

Example of a complete mission setup:

Mission Setup

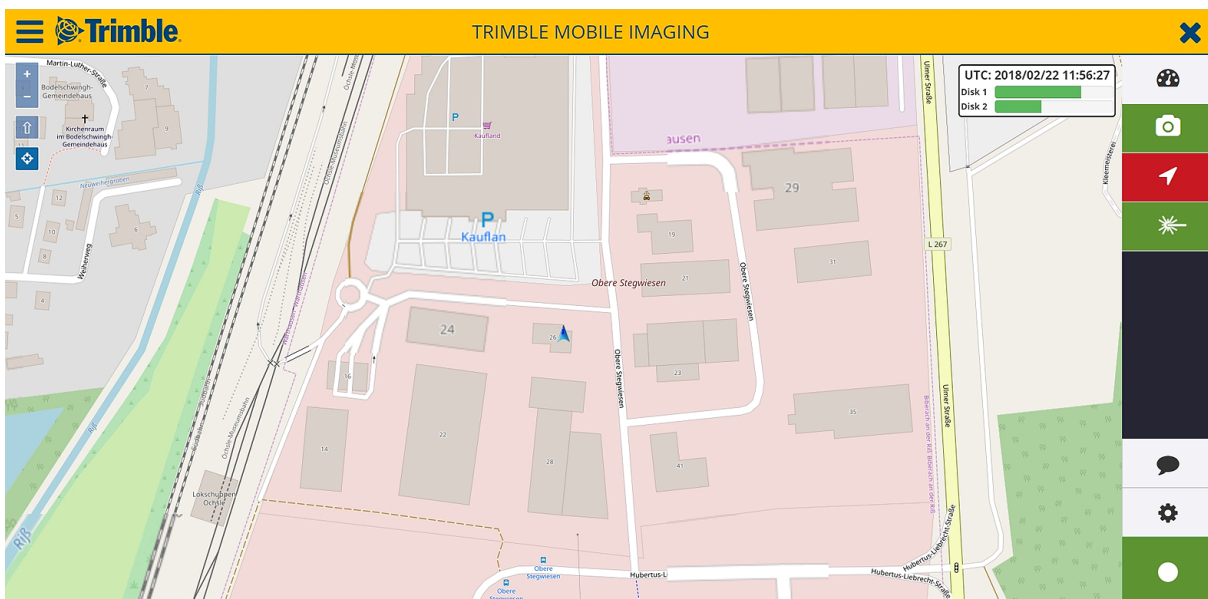
Mission ID	TMX9117070101-000889
Mission Name	Test 2018-02-21-01
Mission Description	Test 2018-02-21 Mission 01
Vehicle	Default Vehicle
Area Name	Area01-a
Area Description	Area01 street order: a
Capture	Urban Area
Cancel Next	

2. Press **Next**. The choices or descriptions you have just made are now listed (see example below):

Mission Start ?	
Mission Name	Test 2018-02-21-01
Mission Description	Test 2018-02-21 Mission 01
Mission Vehicle	Default Vehicle
Area Name	Area01-a
Area Description	Area01 street order: a
Mission Capture	Urban Area
Cancel Start	

3. Confirm these choices or descriptions by pressing **Start**.

The screen looks like this right after the mission has started:



In this screen example:

- The cameras are all time-synchronized (the Camera View button has already turned from orange to green).
- A navigation position is available (a blue arrow is visible on the map).
- Auto-zoom has centered the world map around your current position (provided the system was connected to the Internet and on-line maps were available).

CAUTION –

- Navigation data logging starts automatically as soon as a valid position is available and the blue arrow is visible on the main map window (even if the Navigation View button is still red).
- Navigation data logging is NOT affected by the status color changing from orange to green or vice versa.
- Navigation data logging stops immediately as soon as a mission is closed.
- Laser and image data recording is not possible as long as the Navigation View button stays red. But you may start recording as soon as this button turns from red to orange.
- You can check the mission parameters that are currently used via the TMI menu (mission information).

Changing Capture Settings During a Mission

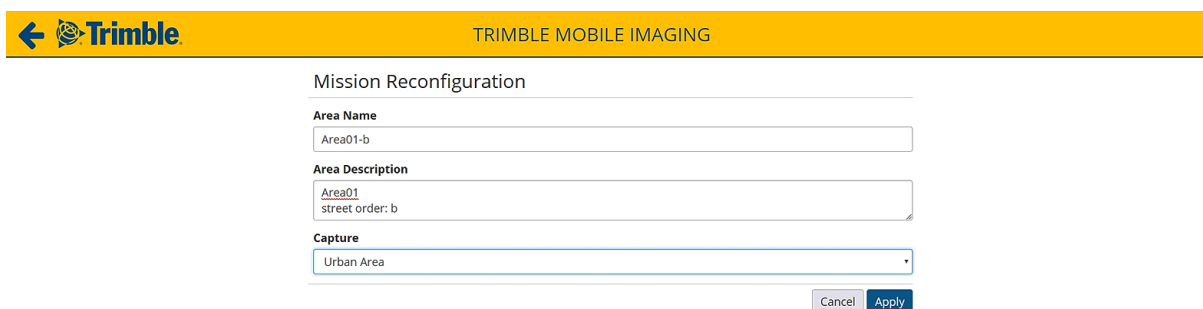
During a mission, you can change the capture settings between two sequences of data collection. This may be useful if for example one or more of your working areas, located at a short distance from each other, require that different capture settings be used.

In this case, you can collect navigation data without any interruption from an area to another, while your capture settings can be changed when switching from an area to the other, so you can collect laser/camera data using different parameters.

By doing this, you will save time as no additional GNSS/IMU initialization procedure will be needed between the moment you end a data collection sequence and the moment you start a new one.

While a mission is in progress, do the following to change the capture settings:

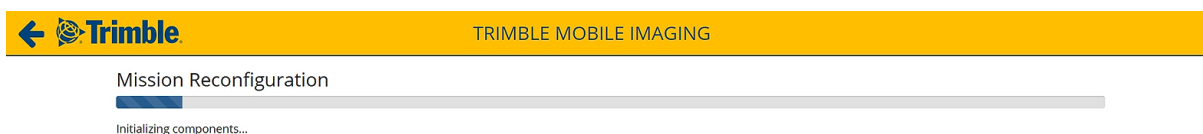
1. Press . The following window then appears from which you can quickly select another preset capture configuration:



2. Select a different capture configuration from the **Capture** drop-down list.

NOTE – The vehicle settings CANNOT be changed in this context as you are still operating in the same vehicle with the same installation parameters.

3. Press **Next** to continue. The following is now displayed:



All sensors are shut down and then re-started with the new configuration.

IMPORTANT! Navigation data logging is kept running while doing this change, and is still running.

Closing a Mission

To close a mission:

- Press . The following message is displayed:

Complete Mission
Do you really want to exit the current mission?
Cancel Complete Mission

- Confirm by pressing the **Complete Mission** button. A progress bar is then displayed indicating that the mission is being closed:

Trimble TRIMBLE MOBILE IMAGING Complete Mission Stopping components... Wait for all components to complete...
Mission will be completed. Please wait.

CAUTION – Navigation data logging stops as soon as you close a mission.

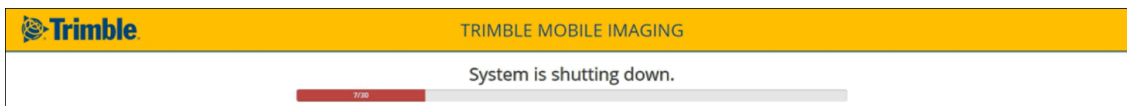
Shutting Down the System

To shut down the system:

1. Press . The following message is then displayed:

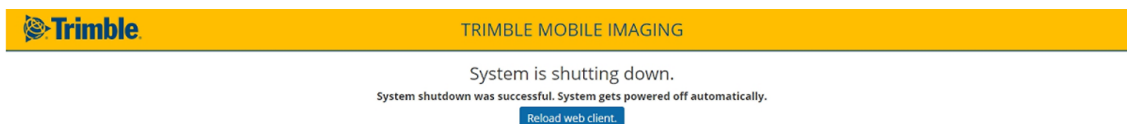


2. Confirm by pressing **Shutdown**. A progress bar is then displayed indicating that the system is being shut down:



NOTE – During shutdown, a system log file is automatically saved to removable Disk 1. This log file can be sent to Trimble Imaging Support for troubleshooting purposes. For more information on how to download log files, see [Log Download, page 50](#). For contact details, see [Support, page 55](#).

At the end of a successful shutdown procedure, you need to re-load your web-client:



Data Structure on SSD (MX scan system)

The data collected by the MX scan system is stored on two SSD drives:

- **SSD1**, the upper one located inside the control unit, contains:
 - Mission database files
 - Mission log file
 - Mission Report
 - Navigation data
 - Laser data from Lasers No. 1 and No. 2
 - Panoramic camera data (camera 4)
 - MX90/MX9 mission database files
- **SSD2**, the lower one inside the control unit, contains oblique camera data (from camera 1, camera 2 and camera 3).

The directory name of each mission is defined by the serial number of the MX scan system and a unique number resulting from an increasing mission counter. The previously entered mission name and area name are saved as attributes into the mission database file.

Example of directory name: TMX9117070101-000759 - MissionName

System:	Trimble MX9 (TMX9)
SN:	117070101
Mission ID:	000759
mission name	MissionName

Data Structure on Backup Data (MX7 only)

The data collected by the MX7 is stored on an internal drive.

The data can be copied to an external drive, which you will connect to the USB3 Port on the MX7 unit.

The backup data contains:

- Mission database files
- Mission log file
- Mission Report
- Navigation data
- Panoramic camera data (camera 4)

The directory name of each mission is defined by the serial number of the MX7 system and a unique number resulting from an increasing mission counter. The previously entered mission name and area name are saved as attributes into the mission database file.

Example of directory name: TMX7117070101-000759 - MissionName

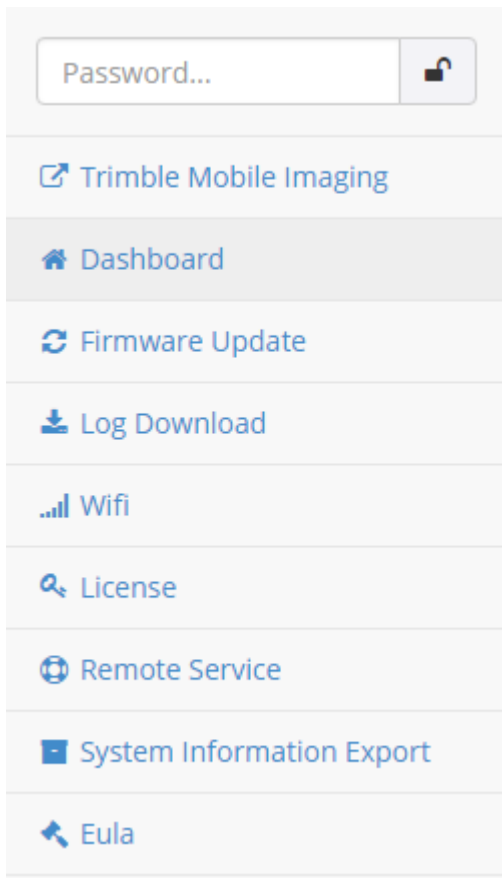
System:	Trimble MX7 (TMX7)
SN:	117070101
Mission ID:	000759
mission name	MissionName

Administration Interface Software

The *Administration Interface* software (TMI.AI) installed on the Trimble Mobile Mapping system is used for maintenance and administration purposes. This software can be accessed from within TMI.Capture, via **Settings> System Administration**, or via:

- MX90:
LAN/WLAN: `http://admin.mx-scan.net`
- MX60:
LAN/WLAN: `http://admin.mx-scan.net`
- MX9:
LAN/WLAN: `http://admin.mx9.net`
- MX50:
LAN/WLAN: `http://admin.mx50.net`
- MX7:
WLAN: `http://192.168.173.1:7076`
LAN: `http://192.168.174.1:7076`

The browser will then show the following TMI.AI menu:



This menu offers the following options:

- **Trimble Mobile Imaging:** Redirects you to the TMI.Capture software.
- **Dashboard:** Allows you to view system information.
- **Firmware Update:** Is used to manage or update the firmware.
- **Log Download:** Is used to download system log files.
- **Wifi:** Is used to access the Wifi settings for an Internet connection.
- **License:** Is used to manage and update warranty licenses.
- **Remote Service:** Is used by Support to start remote sessions.
- **System Information Export:** Generates system information.

Dashboard

The **Dashboard** option gives access to the following information (See screen example below):

- **System Overview**
- **License Information**
- **Firmware Overview**

Dashboard

System Overview	
Manufacturer	Trimble
Model	MX9
SerialNumber	117070101
AssetTag	TMX9117070101
Hardware version	1.0

Firmware Overview	
Version	0.8.2
Description	
Changelog	

License Information	
Status	Valid license
License expire date	2019-02-28
Maintenance expire date	2019-02-15

Firmware Update

The **Firmware Update** option gives access to the following information:

Available Update Files			refresh	
Name	Description	Version		
Linux Update	Linux armstone update	2.0.5		Install
Ladybug6Update	Firmware Update for Ladybug 6	1.23.2		Install
Ladybug5+Update	Firmware Update for Ladybug 5+	1.21.3		Install
X9P Laser Scanner Update	Firmware Update of X9P Laser Scanner	1.4.0.0012		Install
X7P Laser Scanner Update	Firmware Update of X7P Laser Scanner	2.5.12		Install
POSUpdate	Firmware Update of Applanix POS Navigation System11.62	11.62		Install
Network Router Update	Network Router Update	24.10.5		Install
TMI	Trimble Mobile Imaging	5.1.0.73		Install
AP+ Update	Firmware Update of AP+	8.44.014		Install

The **Software Update** view is used to update the system firmware. If a new update is available, you need to copy the update file (which ends with ".tmx.install") to a removable media (USB memory stick).

- With the MX scan system, you need to connect this removable media to the USB1 socket on the MX scan system control unit.
- With the MX7, you need to connect this removable media to the power box.

After pressing the **Refresh** button in this view, you will see the list of all available updates.

If new updates are available, an **Install** button will be shown. The update will start after pressing the **Install** button. The system will first perform a shutdown:

- With the MX scan system, you need to wait until all LEDs on the control unit are off. Then you need to turn the system back on.
- With the MX7, you need to wait at least 2 minutes until the power-off sequence is complete. Then you need to turn the system back on via the power box.

Powering on the system may then take up to 6 minutes, i.e. until the firmware update procedure is complete.

Log Download

This menu option is used to support the system.

Log Download

System Log

refresh

Download locally:

Download

From this screen, you can download the system log information to the tablet or any other device used to access the TMI.AI Administration Interface.

Wifi Setup

This menu option is used to set up the Trimble Mobile Mapping system's internal WiFi device.

MX9 WiFi Access Point

SSID:	Trimble MX9 (117070101)
Password:	12345678 
Country:	Germany 
Connected Clients:	0

Available WiFi Networks


SSID	Signal Strength	Connect
TOOM		
GUEST		
devolo-0e0		
TOOM-ADM		
AndroidAP		

In the **Wifi Access Point** pane, you can see your current Wifi **SSID** and **Password**. This connection is used when you want to connect your user interface device (e.g. tablet or laptop) to the system via Wifi.

The SSID is in the form:

- For MX90: "TrimbleMX90 (117070101)"
- For MX60: "TrimbleMX60 (117070101)"
- For MX9: "TrimbleMX9 (117070101)"
- For MX50: "TrimbleMX50 (117070101)"
- For MX7: "TrimbleMX7 (117070101)"

Where the number inside the brackets is the serial number of the system used.

The password to connect to the Wifi is provided on separate stickers shipped inside the system. This password is unique for each system and cannot be changed.

WARNING – Before you use the system for the first time, you must select your own country so that the Wifi settings appropriate for your country can be applied.

From the **Available Wifi Networks** pane, you can connect to a protected (WPA/WPA2) wireless network. This connection is used when you want to connect the system to the Internet, for example to use online maps within the TMI.Capture software.

NOTE – You will not be able to connect to a Hotspot for which you would need to log in on a web page that requires credentials.

If you want to connect to a Wifi network, you need to press the  button, then the following dialog opens:

Wireless Network Settings

Password:

Cancel

Connect

After the connection to the access point has been established, you will see the following information:

 Internet Connection

SSID:	Flower
Status:	Connected
Channel:	8
Security:	WPA
Signal:	

Disconnect

License

This menu option is used to maintain the system licenses.

License

License Information

Status	Valid license
License expire date	2019-02-28
Maintenance expire date	2019-02-15

Import license from external drive

refresh

There is no newer license

The **License Information** pane provides information about the system license as well as the **Maintenance expire date**. New licenses can be uploaded from this pane. The license file should be copied to a USB memory stick.

- With the MX scan system, the USB memory stick needs to be connected to the USB 1 socket on the MX scan system control unit.
- With the MX7, the USB memory stick needs to be connected to the power box.

Once the USB memory stick is connected, a button should show up in the **Import license from external drive** pane. If it does not, press the **refresh** button. Then press the **Import license from external drive** button to update the system license.

Remote Service

This menu option is used to set up a remote service connection.

RemoteService

Remote Session

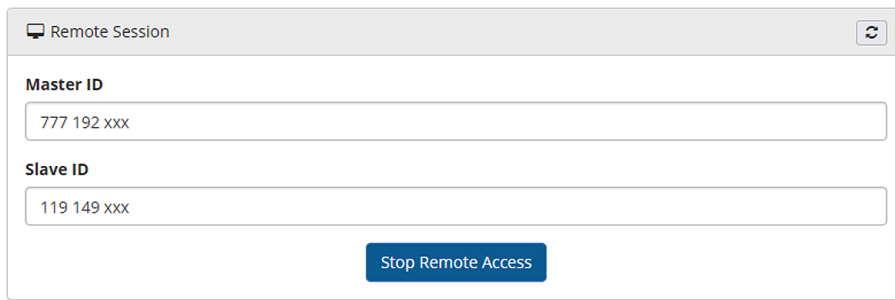
refresh

Start Remote Access

This function provides remote access to the system. If you are facing a problem, Support may need a remote connection to the system.

CAUTION – To start Remote Access, the system should currently be connected to a wireless network that provides access to the Internet.

After pressing the **Start Remote Access** button, it may take some time before the two ID fields below show up:



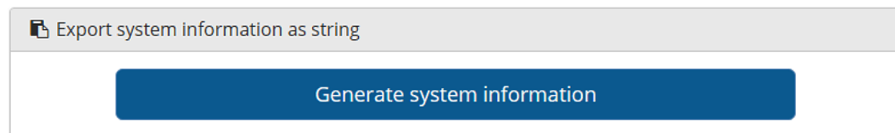
The 'Remote Session' window has a title bar with a computer icon and the text 'Remote Session'. It contains two input fields: 'Master ID' with the value '777 192 xxx' and 'Slave ID' with the value '119 149 xxx'. A blue button labeled 'Stop Remote Access' is positioned below the input fields.

Support will need these IDs to connect to the system.

CAUTION – When remote control of the system is given to Trimble Support, be aware all data on the system are visible to Trimble Support. If there are confidential data on the internal disk, please remove them from the system before granting remote access

System Information Export

This menu option is used to generate a system information file.



The 'Export system information as string' window has a title bar with a document icon and the text 'Export system information as string'. It contains a single blue button labeled 'Generate system information'.

This view gives the ability to export a System Information file, which might be requested by Support.

Support

In case you need support for the Trimble MX system or Trimble Mobile Mapping software, please contact our Customer Support team via email:

Trimble Support Email **mx_support@trimble.com**

Or call:

Support Americas: +1-289-695-4416

Support APAC: +86-105-603-4179

Support Europe & Rest of World: +49 7351 474-02 37

Please describe the issue you are facing as precisely as possible:

- Short description of the problem
- Workflow and how to reproduce the problem
- If occurring during a mission:
 - Location of the mission.
 - Environmental conditions.

Also send the following information about your Trimble MX system:

- Serial Number
- System Log file
- Total operating hours of your Trimble MX system since purchase.
- Photos or videos may also be useful to report a problem.

External Reference Point

Please refer to the manual specific to the MX system used to learn how to measure the external reference point of your system.

Trimble Mobile Mapping Download Center

For system software updates, FAQ, and documentation, go to <https://geospatial.trimble.com/products-and-solutions/mobile-mapping-solutions>, select your Mobile Mapping product and then scroll down to the **Support & Downloads** tile.

